

Ans 5.

(a)	T. ID	Items
	1	bananas, carrots, figs
	2	apples, donuts
	3	bananas, carrots, donuts, figs
	4	carrots, eggs
	5	apples, eggs
	6	donuts,
	7	apples, bananas, carrots
	8	apples, carrots, eggs
	9	carrots, donuts, eggs, figs
	10	apples, donuts, figs

items $\Rightarrow$ apples, bananas, carrots,
donuts, figs, eggs

Support threshold = 0.3

$$N = 10$$

$$\rightarrow s(\text{apples}) = \frac{\sigma}{N} = \frac{5}{10} = 0.5 \{ > 0.3 \} \checkmark$$

$\rightarrow$

$$S(\text{bananas}) = \frac{3}{10} = 0.3 \{= 0.3\} \checkmark$$

$$S(\text{carrots}) = \frac{5}{10} = 0.5 \{> 0.3\} \checkmark$$

$$S(\text{donuts}) = \frac{5}{10} = 0.5 \{> 0.3\} \checkmark$$

$$S(\text{figs}) = \frac{4}{10} = 0.4 \{> 0.3\} \checkmark$$

$$S(\text{eggs}) = \frac{4}{10} = 0.4 \{> 0.3\} \checkmark$$

Thus, for set size = 1,
freq. sets :- all

⊛ Now, for set size = 2 :-

$$S(\text{apples, bananas}) = \frac{\sigma(\text{apples, bananas})}{10} \\ = \frac{1}{10} = 0.1 \{< 0.3\} \times$$

$$S(\text{apples, carrots}) = \frac{2}{10} = 0.2 \{< 0.3\} \times$$

$$S(\text{apples, donuts}) = \frac{2}{10} = 0.2 \{< 0.3\} \times$$

$$S(\text{apples}, \text{eggs}) = \frac{2}{10} = 0.2 \{ < 0.3 \} \times$$

$$S(\text{apples}, \text{figs}) = \frac{1}{10} = 0.1 \{ < 0.3 \} \times$$

$$S(\text{bananas}, \text{carrots}) = \frac{3}{10} = 0.3 \{ = 0.3 \} \checkmark$$

$$S(\text{bananas}, \text{donuts}) = \frac{1}{10} = 0.1 \{ < 0.3 \} \times$$

$$S(\text{bananas}, \text{eggs}) = \frac{0}{10} = 0 \{ < 0.3 \} \times$$

$$S(\text{bananas}, \text{figs}) = \frac{2}{10} = 0.2 \{ < 0.3 \} \times$$

$$S(\text{carrots}, \text{donuts}) = \frac{2}{10} = 0.2 \{ < 0.3 \} \times$$

$$S(\text{carrots}, \text{eggs}) = \frac{3}{10} = 0.3 \{ = 0.3 \} \checkmark$$

$$S(\text{carrots}, \text{figs}) = \frac{3}{10} = 0.3 \{ = 0.3 \} \checkmark$$

$$S(\text{donuts}, \text{figs}) = \frac{3}{10} = 0.3 \{ = 0.3 \} \checkmark$$

$$S(\text{donuts}, \text{eggs}) = \frac{1}{10} = 0.1 \{ < 0.3 \} \times$$

$$S(\text{egg}, \text{fig}) = \frac{1}{10} = 0.1 \{ < 0.3 \} \times$$

$\therefore$ 2 itemset freq. sets are :-
 $\{ \text{bananas}, \text{carrots} \}, \{ \text{carrots}, \text{eggs} \}$
 $\{ \text{donuts}, \text{figs} \}, \{ \text{carrots}, \text{figs} \}$

(*) for 3-itemset freq. set,
 we just have $\{ \text{carrots}, \text{eggs}, \text{figs} \}$
 $(\{ \text{carrots}, \text{egg} \} \times \{ \text{carrots}, \text{figs} \})$

the 2-subsets are not freq.
 so, it will fail anyways.

Hence,

none

(c.) support = 0.3 $\quad c = 0.7$

✓ $(\text{bananas} \rightarrow \text{carrots}) \Rightarrow s(b, c) = 0.3$
 $c = \frac{0.3}{0.3} = 1$

✓ $(\text{eggs} \rightarrow \text{carrots}) \Rightarrow s = 0.3$
 $c = \frac{0.3}{0.4} = 0.75$

These satisfy the condⁿ

$s \geq 0.3$ & $c \geq 0.7$.

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Am 6: TID

Items

1

apples

2

apples, carrots

3

apples, carrots, donuts

4

apples, bananas, eggs

5

apples, bananas, carrots, donuts

6

bananas, donuts, eggs

7

bananas, carrots, figs

8

apples, bananas, carrots

9

apples, bananas, donuts, eggs

10

apples, carrots, eggs

Sorting them:

apples (8) > bananas (6) > carrots (1) >

donuts (4) > eggs (4) > figs (1)



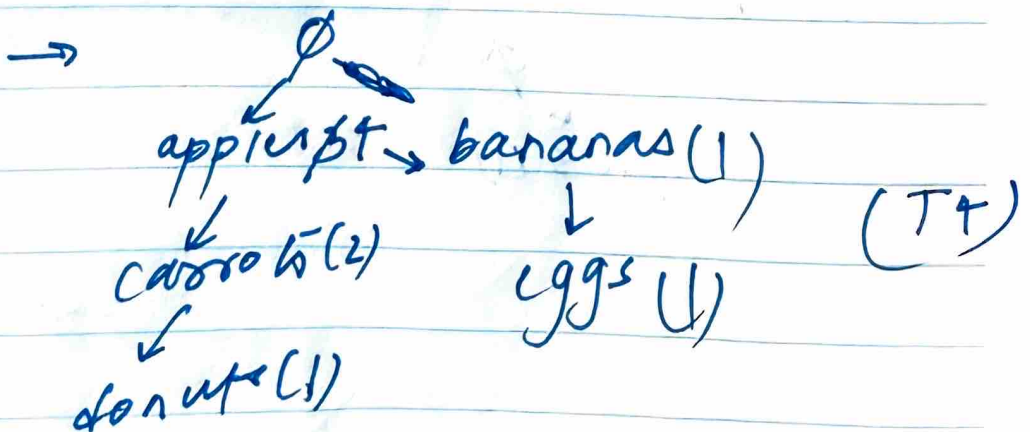
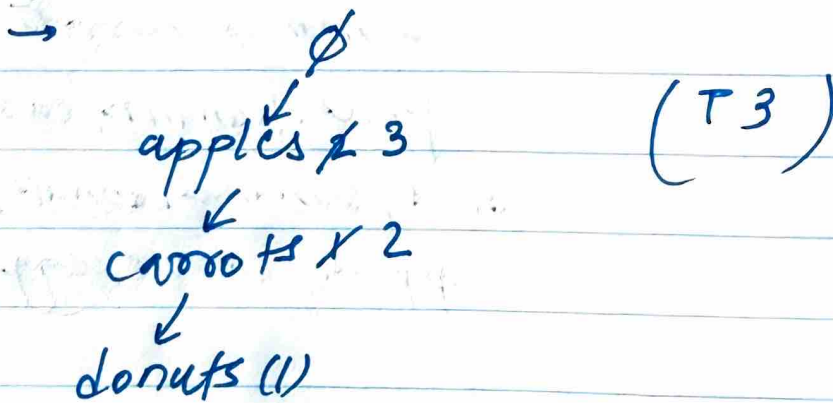
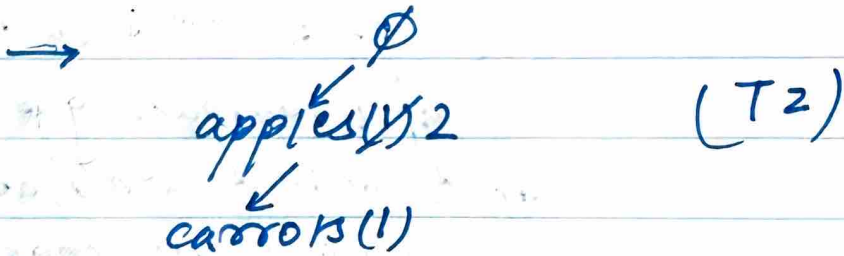
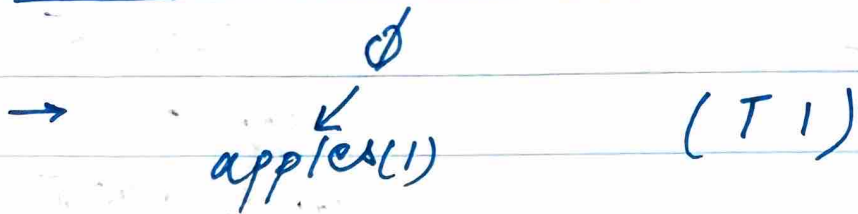
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TID	Itemssets
1	apple
2	apples, carrots
3	apples, carrots, donuts
4	apples, bananas, eggs
5	apples, bananas, carrots, donuts
6	bananas, donuts, eggs
7	bananas, carrots
8	apples, bananas, carrots
9	apples, bananas, donuts, eggs
10	apples, carrots, eggs

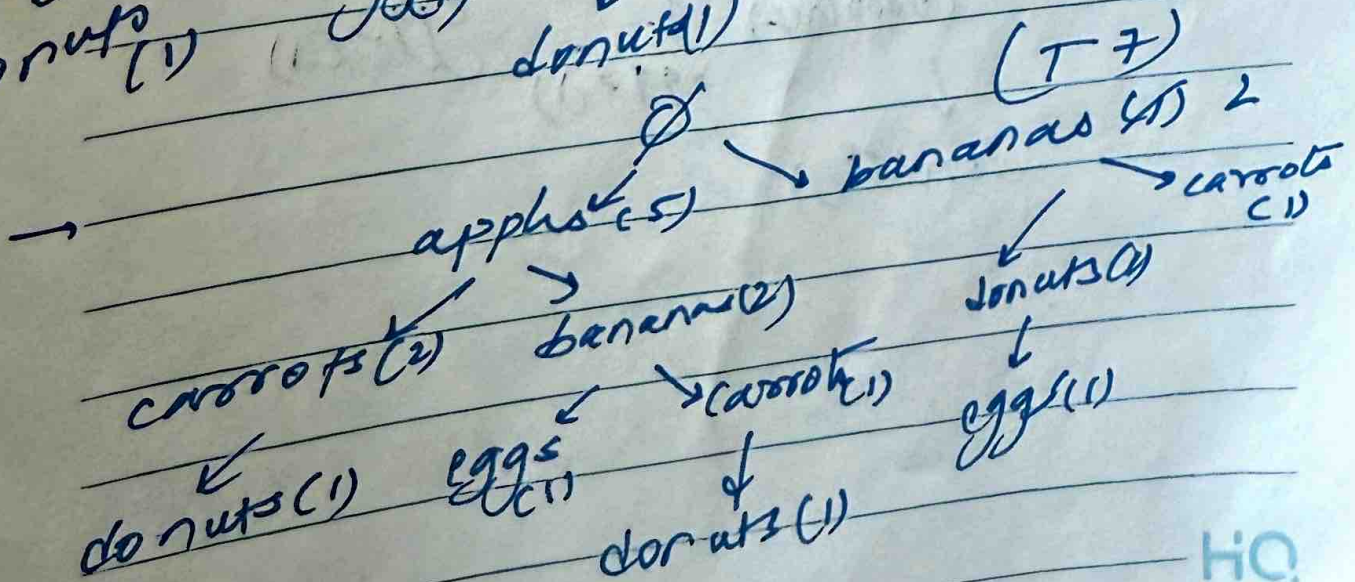
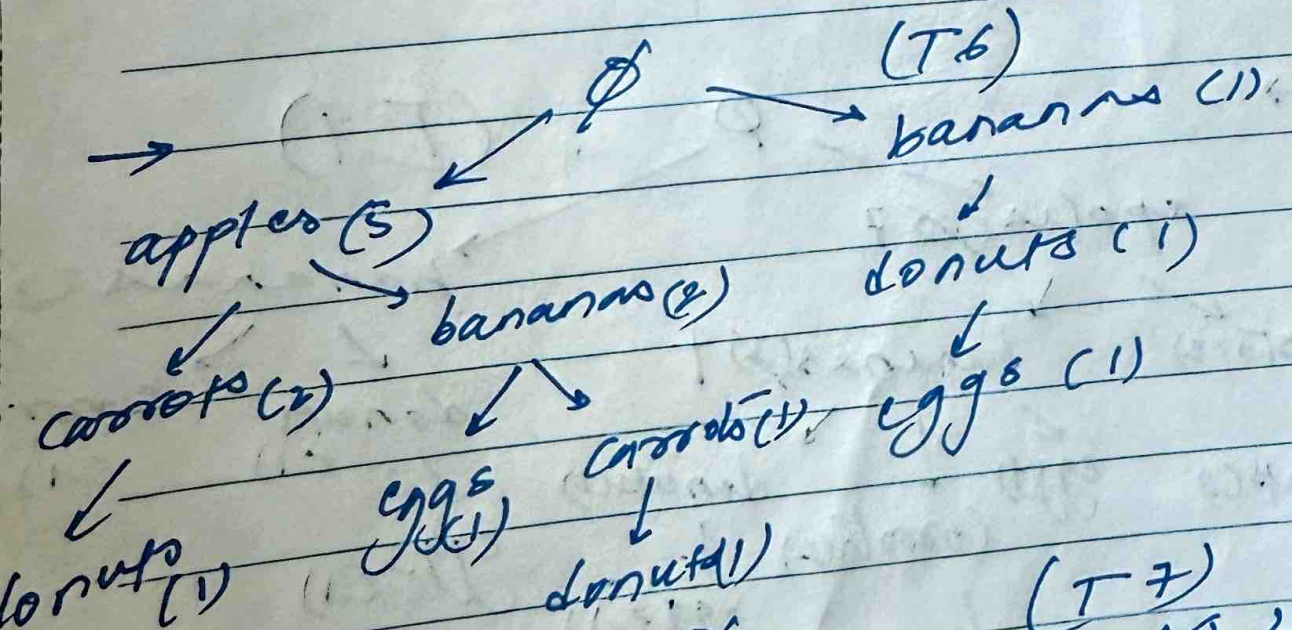
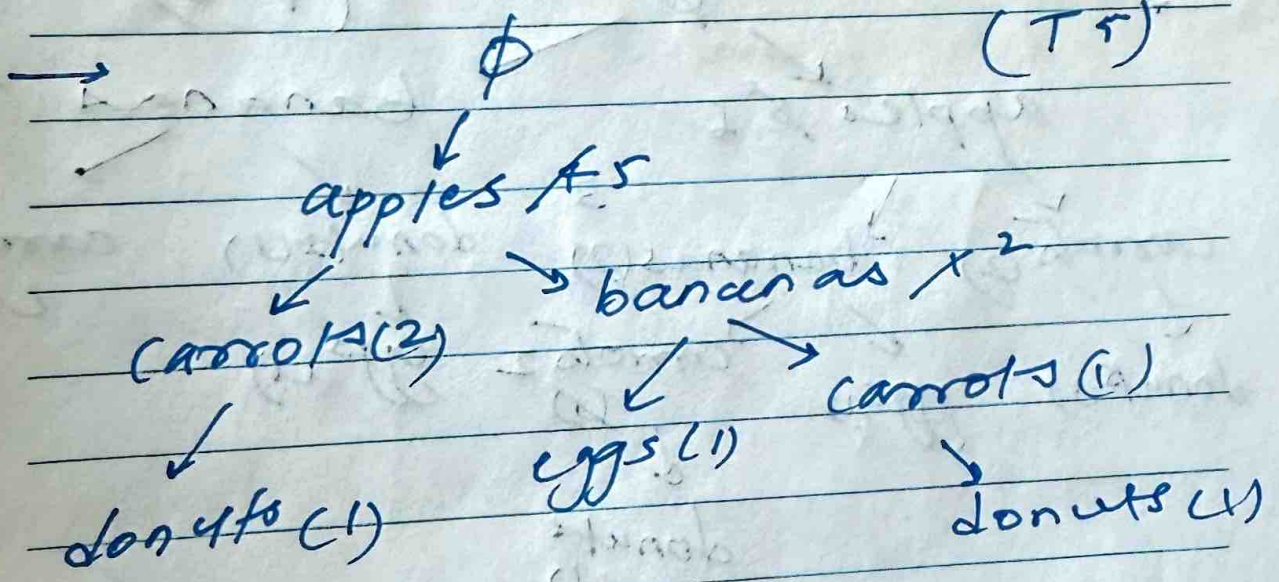
F-p Tree :-

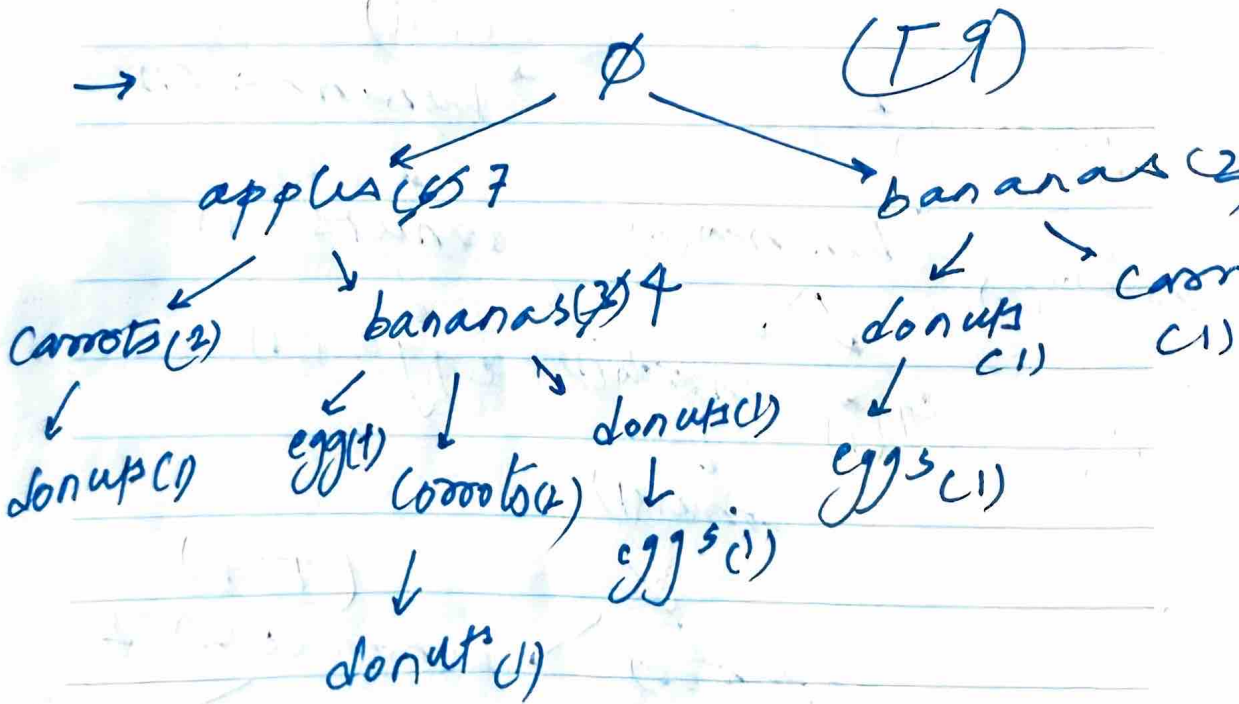
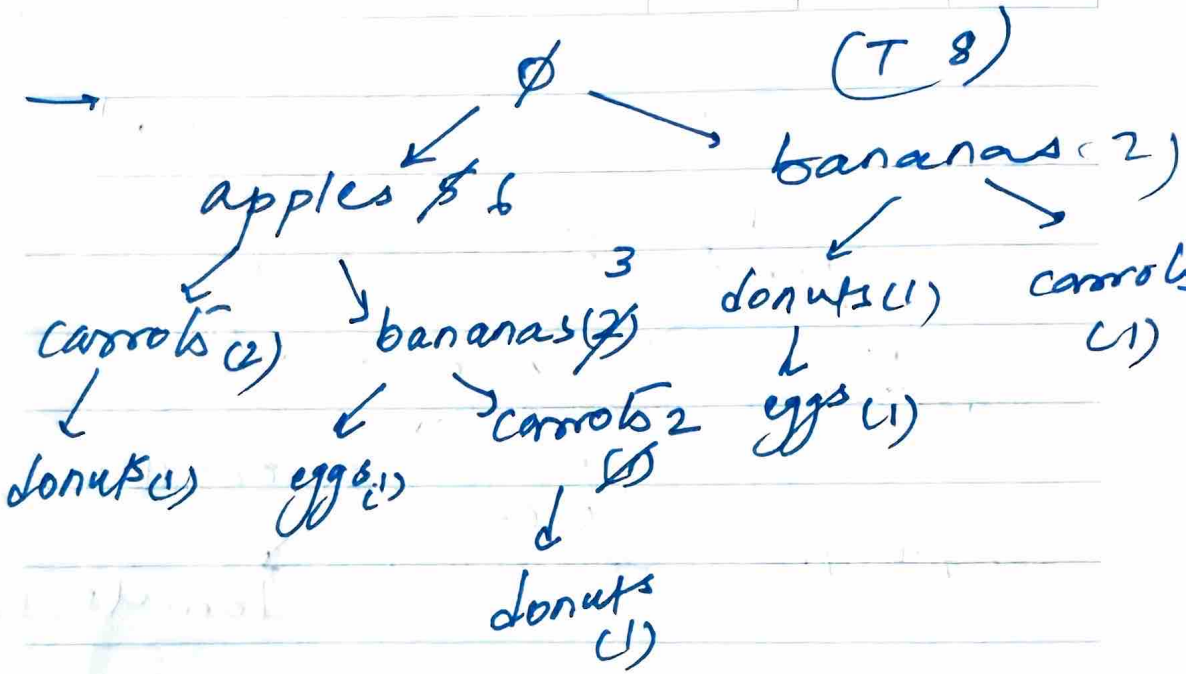
ϕ
 $\swarrow$
 apples (1)



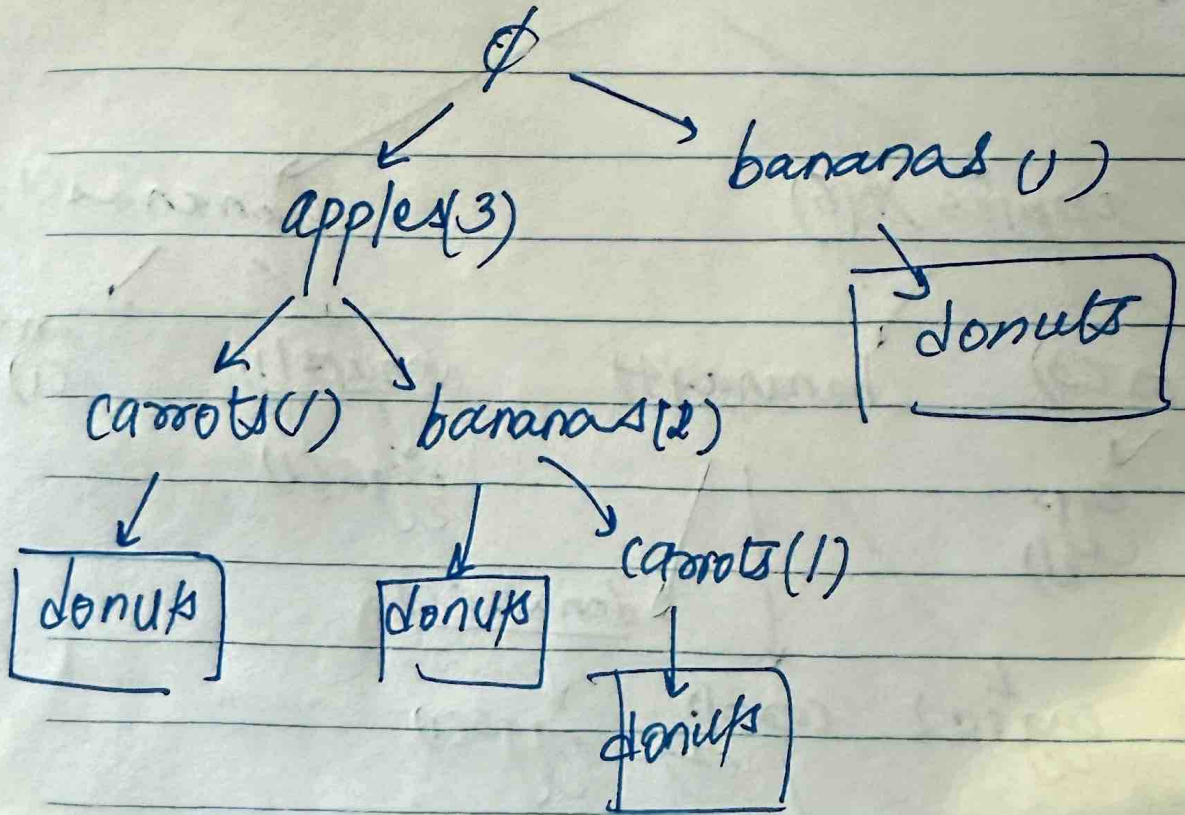
FP Tree:-

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d- conditional Tree:-



freq. itemsets ending with donuts:-

- { apples, donuts } = 3 ✓
 - { bananas, donuts } = 3 ✓
 - { carrots, donuts } = 2 ✓
 - { apples, carrots, donuts } = 2 ✓
 - { apples, bananas, donuts } = 2 ✓
 - { bananas, carrots, donuts } = 1 X
 - { apples, bananas, carrots, donuts } = 1 X
- } Ans.